Sorting Four-Sided Shapes by What They Share

Answer Key

Grades 2–3 Mathematics

Quick Answers

1. 2	4. 20, 3	7. 3	10. —
2. 2	5. —	8. 3	
3. 2	6. 12, 4	9. 2	

Curriculum

Level: Grades 2–3 Mathematics
3.G.A – *problems 1, 2, 3, 5, 6, 7, 8, 9, 10*
3.MD – *problems 4, 6*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

8. *If they got 4: counted the trapezoid, which has only one pair of parallel sides*

1. Circle the cards with four square corners (A square, B long rectangle, C rhombus, D trapezoid).

Test each corner against the corner of a sheet of paper. Card A is a square: all four corners fit the paper corner exactly. Card B is a rectangle: its corners are square too, even though two sides are longer than the other two — side length has nothing to do with the corner test. Card C leans over, so two of its corners are pinched narrower than the paper corner and two are opened wider. Card D has one pair of sides that slant, so only its two bottom corners could be tested and neither is square. Circle A and B. $\boxed{2}$

2. Circle the cards whose four sides are all the same length (A rhombus, B rectangle, C square, D trapezoid).

Measure or compare the sides of each card. Card A leans, but every side is the same length, so it belongs. Card B has two long sides and two short sides, so it does not. Card C is a square: four equal sides, so it belongs. Card D has a long bottom, a shorter top, and two slanted sides of a third length. Circle A and C. $\boxed{2}$

Notice that this group is not the same group as problem 1: card A is in this group and not in that one, and card B is in that group and not this one.

3. Circle the cards with exactly one pair of parallel sides (A parallelogram, B trapezoid, C right trapezoid, D square).

Look at each pair of opposite sides and ask whether they run the same way. Card A has *two* pairs that do, so it has too many — the box asks for exactly one pair. Card B has a top and a bottom that run the same way, while its two slanted sides lean toward each other, so it has exactly one pair. Card C is the same story with one side standing straight up: top and bottom parallel, the other two not. Card D, a square, again has two pairs. Circle B and C. $\boxed{2}$

4. Read the shape-sort graph.

(a) Each bar is labelled with its own number, so add the four groups instead of counting bars: 6+5+6+3. Add the two sixes first to make 12, then $12+5 = 17$ and $17+3 = 20$. $\boxed{20}$

(b) "How many more" means subtract the smaller bar from the bigger one: the "four square corners" bar is at 6 and the "no parallel sides" bar is at 3, so $6 - 3$. $\boxed{3}$

5. How can one card be a rectangle and a square at the same time? (Open response — judge by reading.)

Model answer: a shape goes in the rectangle group when it has four square corners. This card has four square corners, so Rosa is right. A shape goes in the square group when it has four square corners *and* its four sides are all the same length. This card has both, so Ben is right too. The square group is a smaller group living inside the rectangle group: every square is a rectangle, but a rectangle with two long sides is not a square.

Look for: the student names an attribute (corners, sides) as the reason, rather than only saying "it looks like both."

6. Read the line plot of square corners.

(a) Count every $\times$ on the plot, not the numbers underneath: $3 + 2 + 2 + 0 + 5$. That is $3 + 2 = 5$, then $5 + 2 = 7$, then $7 + 5$. $\boxed{12}$

(b) The tallest stack of $\times$ marks sits above one number, and that number is the one that happened most often. The stack above it holds five marks, more than any other stack. $\boxed{4}$

(c) (Open response — judge by reading.) Model answer: try to build one. Draw three square corners; each square corner forces the next side to turn the same way as the corner of a piece of paper, and after three such turns the two ends of the shape close up so that the last corner is square as well. So a four-sided shape that has three square corners always has four. There is no way to stop at exactly three, which is why the mark for three has no $\times$ above it.

7. Nia's group: all four sides the same length (A square, B rectangle, C rhombus, D tilted square).

Compare the four sides of each card. Card A is a square, so all four sides match. Card B is a rectangle with a long pair and a short pair, so it is out. Card C leans but its four sides are equal, so it is in. Card D is a square standing on its point — turning a shape does not change its side lengths, so its four sides are still equal and it is in. Circle A, C and D. $\boxed{3}$

8. Kim's "two pairs of parallel sides" box (A parallelogram, B square, C trapezoid, D rhombus).

Kim counted every card, 4 in all. Check them one at a time. Card A: both pairs of opposite sides run the same way, so it belongs. Card B, a square: both pairs run the same way, so it belongs. Card C is a trapezoid — its top and bottom run the same way, but the two slanted sides lean toward each other and would meet if they kept going, so that is only *one* pair, and this is the card Kim should take back out. Card D, a rhombus, has both pairs parallel, so it belongs. $\boxed{3}$

The mistake to name: Kim saw one pair of parallel sides on card C and stopped checking. The box asks for two pairs, so both pairs have to be tested.

9. Cards with four square corners *and* four equal sides (A square, B rectangle, C rhombus, D tilted square).

This asks for cards that pass *both* tests. Card A passes both: square corners and equal sides. Card B passes the corner test but fails the side test. Card C passes the side test but fails the corner test. Card D is a square turned onto its point: it passes both tests, because turning a card changes neither its corners nor its side lengths. Circle A and D. $\boxed{2}$

The cards that pass both tests are exactly the squares.

10. Draw and compare. (Open response — judge by reading.)

Model answer: a shape with exactly one pair of parallel sides is a trapezoid, for example a shape with a long bottom side, a shorter top side directly above it, and two slanted sides joining them. A shape with two pairs is a parallelogram, a rectangle, a rhombus, or a square — for example a rectangle two dots tall and four dots wide.

A correct shared attribute: both drawings have four straight sides, both have four corners, and both have at least one pair of parallel sides.

Look for: two closed four-sided figures drawn with a straightedge, the parallel pairs genuinely running the same way on the dot grid, and one sentence naming a shared attribute rather than a difference.